Near-side azimuthal and pseudo-rapidity correlations of neutral strange baryons and mesons in d+Au, Cu+Cu and Au+Au collisions at $\sqrt{s_{NN}} = 200$ GeV

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(Dated: March 31, 2015)

We present results on the near-side of high-p_T triggered di-hadron correlations of neutral strange baryons (Λ, $\bar{\Lambda}$) and mesons (K^0_S) at intermediate p_T = 2-6 GeV/c in d+Au, Cu+Cu and Au+Au collisions at $\sqrt{s_{NN}} = 200$ GeV measured by the STAR experiment at RHIC. The near-side di-hadron correlation in ($\Delta\phi$, $\Delta\eta$) contains two structures, a peak which is narrow in $\Delta\phi$ and $\Delta\eta$ consistent with correlations due to jet fragmentation and correlation in $\Delta\phi$ which is broad in $\Delta\eta$ which is attributed dominantly to flow. The dependence of the conditional yield of the jet-like correlation on the trigger particle momentum, associated particle momentum, and centrality are presented. In addition the particle composition of the jet-like correlation is determined using identified associated particles. Data on identified triggers are qualitatively described by PYTHIA, however the composition of the jet-like correlation is not quantitatively described by PYTHIA.

PACS numbers: 25.75.-qRelativistic heavy-ion collisions 25.75.GzParticle correlations and fluctuations

I. INTRODUCTION

Ultra-relativistic heavy-ion collisions create a unique environment for the investigation of nuclear matter at extreme temperatures and energy densities in the laboratory. Measurements of nuclear modification factors [1–4] have shown that the nuclear medium created is opaque to particles with large transverse momentum (p_T). Moreover, anisotropic elliptic flow measurements demonstrate that the medium exhibits partonic degrees of freedom and has properties close to those expected of a perfect fluid [5, 6].

Studies of jets in heavy ion collisions are possible through single particle measurements [1–4], di-hadron correlations [7–17], and measurements of reconstructed jets [3, 18–21]. Measurements of reconstructed jets are useful for studying higher momenta and have provided direct evidence for partonic energy loss in the medium. However, di-hadron correlations provide complementary measurements since they enable studies at intermediate momenta, where the interplay between jets and the medium is important and direct jet reconstruction is no longer possible.

Properties of jets have been studied extensively using di-hadron correlations relative to a trigger particle with large transverse momentum at the Relativistic Heavy Ion Collider (RHIC) [7–14] and the Large Hadron Collider (LHC) [15–17]. Systematic studies of associated particle distributions on the opposite side of the trigger particle revealed significant modification, including the disappearance of the peak at intermediate p_T (2 GeV/c) [10, 22] and its reappearance at high p_T [11, 23]. The associated particle distribution on the near side of the trigger particle, the subject of this paper, is also significantly modified in central Au+Au collisions [8, 12, 24]. In p+p and d+Au collisions, there is a peak that is narrow in azimuth ($\Delta\phi$) and pseudorapidity ($\Delta\eta$) around the trigger particle, which we refer to as the jet-like correlation. In Cu+Cu and Au+Au collisions this peak is observed to be broader than that in d+Au collisions, although the yields are comparable [7]. This indicates that the jet-like correlation is dominantly produced through fragmentation.

Besides the shape modifications of jet-like correlations at intermediate transverse momenta, the production mechanism of hadrons may differ from simple fragmentation. In central Au+Au collisions baryon production in the bulk is enhanced relative to that in p+p or d+Au collisions [25, 26]. The baryon to meson ratios measured in Au+Au collisions increase with p_T until reaching a maximum of ≈ 3 relative to p+p collisions at $p_T \approx 3$ GeV/c in both the strange and non-strange quark sectors. A fall-off of the baryon to meson ratio is observed for $p_T > 3$ GeV/c and both the strange and non-strange baryon to meson ratios in Au+Au collisions approach the values measured in p+p collisions at $p_T \approx 6$ GeV/c.

In this paper, studies of two-particle correlations on the near-side in d+Au, Cu+Cu and Au+Au collisions at $\sqrt{s_{NN}} = 200$ GeV measured by the STAR experiment are presented. Results from two-particle correlations in pseudo-rapidity and azimuth for neutral strange baryons (Λ, $\bar{\Lambda}$) and mesons (K^0_S) at intermediate p_T ($2<p_T<6$ GeV/c) in the different collision systems are compared to unidentified charged hadron correlations. Both identified trigger particles associated with charged particles (K^0_S-h, Λ-h) and unidenfitied charged trigger particles associated with identified strange particles (h-K^0_S, h-Λ) are studied. The near-side jet-like yield is studied as a function of centrality of the collision and transverse momentum of trigger and associated particles to look for possible flavor and baryon/meson dependence. The composition of the jet-like correlation is studied using identified associated particles to investigate possible production mechanisms. The results are compared to expectations from PYTHIA.

II. EXPERIMENTAL SETUP AND PARTICLE RECONSTRUCTION

STAR [27] is a multi-purpose spectrometer with a full azimuthal coverage consisting of several detectors inside a large solenoidal magnet with a uniform magnetic field of 0.5 T applied along the beam line. This analysis is based exclusively on charged particle tracks detected and reconstructed in the Time Projection Chamber (TPC) [28] with a pseudo-rapidity acceptance $|\eta| < 1.5$. The TPC has in total 45 pad rows in the radial direction allowing up to 45 independent spatial and energy loss (dE/dx) measurements for each charged particle track. The TPC momentum resolution is determined to be $\Delta p/p \sim 0.0078 + 0.0098 \cdot p_T$ (GeV/c) [28]. Charged particle tracks used in this analysis were required to have at least 15 fit points in the TPC, a DCA to the primary vertex of less than 1 cm and a pseudorapidity $|\eta| < 1.0$. The results presented in this paper are based on analysis of data from d+Au, Cu+Cu, and Au+Au collisions at $\sqrt{s_{NN}} = 200$ GeV taken by the STAR experiment in 2003, 2005, and 2004, respectively.

For d+Au collisions, the events analyzed were selected using a minimally biased (MB) trigger requiring at least one beam-rapidity neutron in the Zero Degree Calorimeter (ZDC), located 18 m from the nominal interaction point in the Au beam direction and accepting 95±3% of the hadronic cross section [29]. For Cu+Cu collisions, the MB trigger was based on the combined signals from the Beam-Beam Counters (BBC) placed at forward pseudo-rapidity ($3.3 < |\eta| < 5.0$) and a coincidence between the two ZDCs. The MB Au+Au events required a coincidence between the two ZDCs, a signal in both BBCs and a minimum charged particle multiplicity in an array of scintillator slats aligned parallel to the beam axis and arranged in a barrel, the Central Trigger Barrel (CTB), to reject non-hadronic interactions. An additional online trigger for central Au+Au collisions was used to sample the most central 12% of the total hadronic cross section. This trigger was based on the energy deposited in the ZDCs in combination with the multiplicity in the CTB.

In order to achieve a more uniform detector acceptance in Cu+Cu and Au+Au data sets, only those events with the primary vertex position along the longitudinal beam direction (z) within 30 cm of the center of the STAR detector were used for the analysis. For the d+Au collisions this vertex position selection was extended to $|z| < 50$ cm. The number of events after the vertex cuts in individual data samples is summarized in Table I.

TABLE I: Number of events after cuts (see text) in the data samples analyzed.

System	Centrality	No. of events [10^6]
d+Au	0-95%	3
Cu+Cu	0-60%	38
Au+Au	0-80%	28
Au+Au	0-12%	17

We identify weakly decaying neutral strange (V^0) particles Λ, $\bar{\Lambda}$ and K^0_S by topological reconstruction of their decay vertices from their charged hadron daughters measured in the TPC as described in [30]:

$$\begin{aligned} \Lambda &\rightarrow p + \pi^-, BR = (63.9 \pm 0.5)\% \\ \bar{\Lambda} &\rightarrow \bar{p} + \pi^+, BR = (63.9 \pm 0.5)\% \\ K^0_S &\rightarrow \pi^+ + \pi^-, BR = (68.95 \pm 0.14)\% \end{aligned} \quad (1)$$

where BR denotes the branching ratio. The STAR V^0 finding reconstruction software pairs oppositely charged particle tracks into V^0 candidates. Reconstructed Λ and K^0_S particles are required to be within $|\eta| < 1.0$. Topological cuts are optimized for each data set and chosen to reach a signal to background ratio of at least 15:1. For the analyses presented here, no difference was observed between results with Λ and $\bar{\Lambda}$ trigger particles. Therefore the correlations were added to increase the statistical significance of the results. In the remainder of the discussion the combined particles are refered to simply as Λ baryons.

III. METHOD

A. Correlation technique

The analysis in this paper follows the method in [7]. A high-p_T trigger particle was selected and the raw distribution of associated tracks relative to that trigger particle in pseudorapidity ($\Delta\eta$) and azimuth ($\Delta\phi$) is formed. This distribution, $d^2N_{\mathrm{raw}}/d\Delta\phi\, d\Delta\eta$, is normalized by the number of trigger particles, N_{trigger}, and corrected for the efficiency and acceptance of associated tracks:

$$\frac{d^2N}{d\Delta\phi\, d\Delta\eta}(\Delta\phi, \Delta\eta) = \frac{1}{N_{\mathrm{trigger}}}\frac{d^2N_{\mathrm{raw}}}{d\Delta\phi d\Delta\eta} \frac{1}{\varepsilon_{\mathrm{assoc}}(\phi,\eta)}\frac{1}{\varepsilon_{\mathrm{pair}}(\Delta\phi,\Delta\eta)}. \quad (2)$$

The efficiency correction $\varepsilon_{\mathrm{assoc}}(\phi,\eta)$ is a correction for the single particle reconstruction efficiency in TPC and $\varepsilon_{\mathrm{pair}}(\Delta\phi,\Delta\eta)$ is a correction for the finite TPC track-pair acceptance in $\Delta\phi$ and $\Delta\eta$. Since the correlations are normalized by the number of trigger particles, the efficiency correction is only applied for the associated particle. The fully corrected correlation functions are averaged between positive and negative $\Delta\phi$ and $\Delta\eta$ regions and are reflected about $\Delta\phi$ =0 and $\Delta\eta$ =0 in the plots.

B. Single charged track efficiency correction

For unidentified charged associated particles, the efficiency correction $\varepsilon_{\mathrm{assoc}}(\phi,\eta)$ is the correction for charged hadrons, identical to that applied in [7]. This single charged track reconstruction efficiency is determined as a function of p_T, η, and centrality by simulating the TPC response to a particle and embedding the simulated signals into a real event. The single charged track efficiency is found to be approximately constant for $p_T > 2$ GeV/c and ranges from around 75% for central Au+Au events to around 85% for peripheral Cu+Cu events. The efficiency for reconstructing a track in d+Au events is 89%.

For identified associated strange particles, the reconstruction efficiency $\varepsilon_{\mathrm{assoc}}(\phi, \eta)$ is determined in a similar way, but forcing the simulated particle to decay through the channel in Equation 2 and then correcting for the respective branching ratio. The efficiency for Λ, $\bar{\Lambda}$, and K^0_S ranges from 8% to 15%, increasing with momentum and decreasing with system size [31]. No correction for the reconstruction efficiency is applied for identified trigger particles because the reconstruction efficiency does not vary significantly within the p_T^{trigger} bins used in this analysis.

The systematic uncertainty associated with the efficiency correction for unidentified associated hadrons, 5%, is strongly correlated across centralities and p_T bins for each data set but not between data sets. For identified associated particle ratios the systematic errors on the efficiency correction partially cancel out and are negligible compared to the statistical uncertainties.

For the inclusive spectra the feeddown correction due to secondary Λbaryons from Ξ baryon decays is 15%, independent of p_T [32]. For identified Λ trigger particles, we assume that feeddown lambdas do not change the correlation since little trigger particle type dependence is observed in the data. This is consistent with the available data within errors. For identified associated particles, we assume the same correlation between primary and secondary Λ baryons and correct the yield of Λ associated particles by reducing the yield by 15%.

C. Pair acceptance correction

The requirement that each track falls within $|\eta| < 1.0$ in TPC results in a limited acceptance for track pairs. The geometric acceptance for a track pair is $\approx$ 100% for $\Delta\eta \approx 0$ and close to 0% near $\Delta\eta \approx 2$. The track pair acceptance is limited in azimuth by the 12 TPC sector boundaries, leading to dips in the acceptance of track pairs in $\Delta\phi$. To correct for the limited geometric acceptance, a mixed event analysis was performed using trigger particles from one event combined with associated particles from another event, as done in [12]. The event vertices were required to be within 2 cm of each other along the beam axis and the events were required to have the same charged particle multiplicity within 10 particles. To increase statistics of the mixed event sample, each event with a trigger particle was mixed with ten other events.

D. Track merging correction

The track merging effect in unidentified hadron (h) correlations discussed in [7] is also present for V^0-h and h-V^0 correlations. This effect leads to a dip at small $\Delta\phi$ and $\Delta\eta$ due to overlap between the trigger and associated particles. When one of the particles is a V^0, this overlap is between one of the V^0 daughter particles and the unidentified hadron. The size of the dip due to track merging depends strongly on the relative momenta of the particle pair. This effect is larger when the overlapping particles' momenta are closer. For V^0-h correlations, the typical associated particle momentum is approximately 1.5 GeV/c. Since the K^0_S decay is symmetric, this means that the track merging effect is greatest for K^0_S-h correlations with a trigger K^0_S momentum of approximately 3 GeV/c. In a Λ decay, the proton daughter carries more of the Λ momentum than the pion daughter. Therefore this effect is larger for Λ trigger particles with lower momenta. Because track merging affects both signal and background particles and the signal sits on top of a large combinatorial background, the effect is larger for collisions with a higher charged track multiplicity. Since the dip in V^0-h and h-V^0 correlations is the result of a V^0 daughter merged with an unidentified hadron, the dip is wider in $\Delta\phi$ and $\Delta\eta$ than in unidentified hadron correlations. When the track merging dip is present, it is corrected by fitting a Gaussian to the peak, excluding the region impacted by track merging, and using the Gaussian fit to extract the yield.

In the absence of a clear dip from track merging, the yield calculated from bin counting is consistent with the yield from the Gaussian fit. When the residual track merging effect is corrected by a fit, the track merging correction applied by the fit is approximately the same order as the statistical error on the yield. We therefore conclude that when no dip is evident, the track merging effect is negligible compared to the statistical error on the yield.

E. Yield extraction

The notation and method used to extract the yield in this paper follow [7, 12]. The jet-like correlation is narrow in both $\Delta\phi$ and $\Delta\eta$ and is contained within the kinematic cuts in p_T^{trigger} and $p_T^{\mathrm{associated}}$ used in this analysis. The di-hadron correlation from Equation 2 is projected onto the $\Delta\eta$ axis:

$$\left.\frac{dN}{d\Delta\eta}\right|_{a,b} \equiv \int_a^b d\Delta\phi \frac{d^2N}{d\Delta\phi d\Delta\eta}. \qquad (3)$$

All other correlations, including those from v_2, v_3, and higher order flow harmonics, are assumed to be independent of $\Delta\eta$ within the η acceptance of the analysis, consistent with [12, 33–35]. We make the assumption that

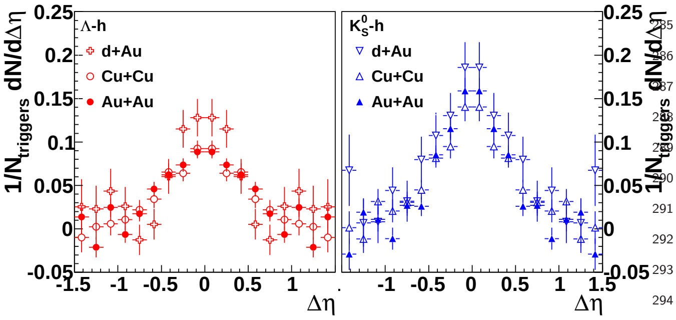

FIG. 1: Corrected correlation functions $\frac{dN_J}{d\Delta\eta}$ in $|\Delta\phi| = 1.0$ for $3 < p_T^{\mathrm{trigger}} < 6$ GeV/c and 1.5 GeV/$c < p_T^{\mathrm{associated}} < p_T^{\mathrm{trigger}}$ for Λ-h (left) and K^0_S-h (right) for minimum bias d+Au, 0-20% Cu+Cu, and 40-80% Au+Au collisions at $\sqrt{s_{NN}} = 200$ GeV after background subtraction. The data have been reflected about $\Delta\eta = 0$.

the η dependence observed for v_3 measured using the two particle cumulant method [36] is entirely due to nonflow. This method explicitly subtracts such correlated backgrounds in addition to uncorrelated background. This means that the jet-like yield can be determined by:

$$\frac{dN_J}{d\Delta\eta}(\Delta\eta) = \left.\frac{dN}{d\Delta\eta}\right|_{-0.78,0.78} - b_{\Delta\eta} \tag{4}$$

where $b_{\Delta\eta}$ is a constant offset determined by fitting a constant background $b_{\Delta\eta}$ plus a Gaussian to $\frac{dN_J}{d\Delta\eta}(\Delta\eta)$. Variations in the method for extracting the constant background, such as fitting a constant at large $\Delta\eta$, lead to differences in the yield smaller than the statistical error due to the background alone. Sample correlations are given in Figure 1. When the dip due to track merging is negligible, the yield determined from fit is discarded to avoid any assumptions about the shape of the peak and instead we integrate Equation 4 over $\Delta\eta$ using bin counting to determine the jet-like yield $Y_J^{\Delta\eta}$:

$$Y_J^{\Delta\eta} = \int_{-0.78}^{0.78} d\Delta\eta\, \frac{dN_J}{d\Delta\eta}(\Delta\eta)\ . \tag{5}$$

Where track merging is negligible the yield from the fit and from bin counting are consistent. Therefore for data where the dip due to track merging is significant, the yield from the Gaussian fit is used, excluding the region affected by track merging, and the statistical errors from the fit exceed any error from the assumption of a Gaussian shape.

IV. RESULTS

A. Charged-V^0 correlations

Previous studies demonstrated that the jet-like correlation in h-h correlations is independent of collision system [7, 12, 37] and that it is qualitatively described by PYTHIA [7]. This indicates that the jet-like correlation is dominantly produced by fragmentation, even at intermediate momenta ($2 < p_T < 6$ GeV/c) where recombination is expected to contribute substantially to the bulk production of particles. The composition of the jet-like correlation can be studied using correlations with identified associated particles. For the analysis presented here, the statistics of the d+Au data were limited and the Au+Au data set was limited by the presence of residual track merging. Therefore it was only possible to determine the composition of the jet-like correlation in Cu+Cu collisions.

These measurements are compared to inclusive baryon to meson ratios in p+p collisions from STAR [38] and ALICE [39] and simulations in p+p collisions in PYTHIA [40] using the Perugia 2010 and 2011 [41] tunes in Figure 2. The ratio in Cu+Cu collisions is consistent with the inclusive particle ratios from p+p. This further supports earlier observations that the jet-like correlation is dominantly produced by fragmentation since the production of particles both in p+p collisions at these momenta and in the jet-like correlation are expected to be dominated by fragmentation. This implies that production of strange particles through recombination is not significant in the jet-like correlation, even in A+A collisions where the inclusive spectra show an enhancement of Λ of up to a factor of three relative to the K^0_S. The inclusive $\frac{\Lambda+\bar{\Lambda}}{2K^0_S}$ ratio is also consistent between different energies.

However, the production of strangeness and of baryons is potentially more complicated. PYTHIA is able to describe π and ω production [42, 43], however, generally underestimates production of strange particles, including the K^0_S, and baryons, including the Λ [38, 39, 42, 43]. As shown in Figure 2 PYTHIA underestimates the $\frac{\Lambda+\bar{\Lambda}}{2K^0_S}$ ratio, both in Tune A and the Perugia tunes, which improved the description of inclusive strangeness production at LHC energies. PYTHIA also underestimates the $\frac{\Lambda+\bar{\Lambda}}{2K^0_S}$ ratio at $\sqrt{s}$ = 900 and 7000 GeV and predicts a change in the $\frac{\Lambda+\bar{\Lambda}}{2K^0_S}$ ratio with energy. For each of the PYTHIA tunes used, the ratio in the inclusive spectra and the ratio in the jet-like correlation are comparable and lower than that observed in the data. Even though PYTHIA underestimates the overall production of both the Λ baryon and the K^0_S meson, it predicts that the ratios in the inclusive spectra and the jet-like correlation are comparable, as observed in the data.

The h-K^0_S and h-Λ correlations are likely dominated by gluon and light quark jets, indicating that PYTHIA is underestimating the production of strange particles in light quark jets. However, K^0_S-h and Λ-h correlations should have larger contributions from jets containing strange particles. It may be easier for PYTHIA to describe the fragmentation of an initial strange quark than to describe the production of strange particles from non-strange jets.

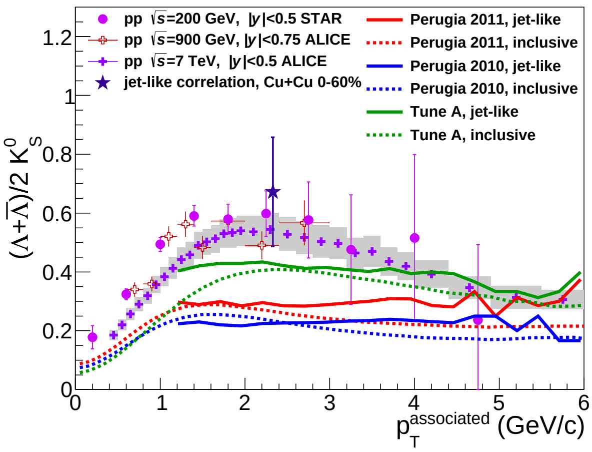

FIG. 2: Λ/K^0_S ratio measured in the jet-like correlation in 0-60% Cu+Cu collisions at $\sqrt{s_{NN}} = 200$ GeV for $3 < p_T^{\rm trigger} < 6$ GeV/c and $2.0 < p_T^{\rm associated} < 3.0$ GeV/c along with this ratio obtained from inclusive p_T spectra in p+p collisions. Data are compared to calculations from PYTHIA [40] using the Perugia 2010 and 2011 tunes [41] and Tune A [44]. The 5% systematic error due to the uncertainty on the associated particle's efficiency is not shown.

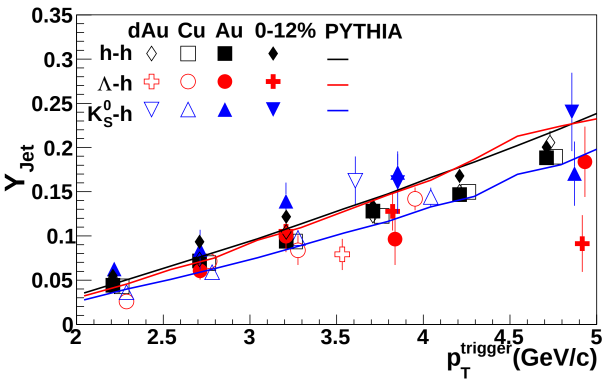

FIG. 3: The jet-like yield in $|\Delta\eta| <0.78$ as a function of $p_T^{\rm trigger}$ for h-h [7], K^0_S-h, and Λ-h correlations for 1.5 GeV/c $< p_T^{\rm associated} < p_T^{\rm trigger}$ in minimum bias d+Au, 0-60% Cu+Cu, and Au+Au collsions at $\sqrt{s_{NN}} = 200$ GeV. The data from Au+Au collisions are shown in two centrality bins, 40-80% and 0-12%. The data are compared to PYTHIA [40] calculations using the Perugia 2011 tune [41]. Data have been offset for visibility. The 5% systematic error due to the uncertainty on the associated particle's efficiency is not shown.

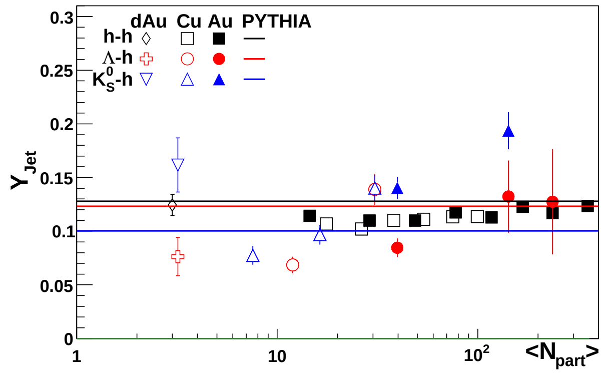

FIG. 4: Centrality dependence of the jet-like yield of h-h, K^0_S-h, and Λ-h correlations for $3 < p_T^{\rm trigger} < 6$ GeV/c and 1.5 GeV/c $< p_T^{\rm associated} < p_T^{\rm trigger}$ in d+Au, Cu+Cu, and Au+Au collisions at $\sqrt{s_{NN}} = 200$ GeV. The data are compared to PYTHIA [40] calculations using the Perugia 2011 tune [41]. Points have been shifted for visibility.The 5% systematic error due to the uncertainty on the associated particle's efficiency is not shown.

B. Correlations with identified strange trigger particles

The jet-like yield as a function of $p_T^{\rm trigger}$ is shown in Figure 3 for h-h [7], K^0_S-h, and Λ-h correlations for d+Au, Cu+Cu, and Au+Au collisions at $\sqrt{s_{NN}} = 200$ GeV. The data are tabulated in Table II. Due to residual track merging effects, fits are used for Λ-h correlations in Cu+Cu collisions with $2.0 < p_T^{\rm trigger} < 3.0$ GeV/c, in 0-12% Au+Au collisions for $3.0 < p_T^{\rm trigger} < 4.5$ GeV/c, and in 40-80% Au+Au collisions for $2.0 < p_T^{\rm trigger} <$ 4.5 GeV/c. There is no significant difference in the yields between the collision systems as observed earlier for h-h correlations in [7]. This includes no significant difference between results from Au+Au collisions in 40-80% and 0-12% central collisions, although there is a hint that the Λ-h jet-like yield may be lower in central Au+Au collisions, particularly at high p_T. The jet-like yield from K^0_S-h correlations is systematically above the jet-like yield from h-h correlations and the jet-like yield from Λ-h is systematically below the jet-like yield from h-h correlations, particularly in Au+Au collisions. Data are compared to PYTHIA calculations from the Perugia 2011 [41] tune. We focus on the Perugia 2011 tune because this is the latest tune incorporating data on strangeness at the LHC and we saw no qualitative difference between various PYTHIA tunes. PYTHIA describes the trend of the jet-like yield well for all particle species. However, the trigger particle type ordering in PYTHIA is opposite that observed in the data, with the Λ-h jet-like yield comparable to the h-h jet-like yield, which is higher than K^0_S-h jet-like yield.

Next the jet-like yields are studied as a function of collision centrality expressed in terms of number of participating nucleons ($N_{\rm part}$) calculated from the Glauber model [45]. The extracted jet-like yield as a function of $N_{\rm part}$ is shown in Figure 4 for h-h [7], K^0_S-h, and Λ-h correlations for d+Au, Cu+Cu, and Au+Au collisions at $\sqrt{s_{NN}} = 200$ GeV. All yields are determined using bin counting. While there is no centrality dependence in the jet-like yield of h-h correlations, there is a centrality dependence in the yields of the K^0_S-h and Λ-h correlations. These data are compared to PYTHIA [40] calculations from the Perugia 2011 [41] tune.

TABLE II: The jet-like yield in $|\Delta\eta\ |$<0.78 as a function of $p_T^{\rm trigger}$ for K_S^0-h, and Λ-h correlations for 1.5 GeV/c < $p_T^{\rm associated}$ < $p_T^{\rm trigger}$ in minimum bias d+Au, 0-60% Cu+Cu, and Au+Au collsions at $\sqrt{s_{NN}}$ = 200 GeV, as shown in Figure 3.

p_T	K_S^0-h MB d+Au	K_S^0-h 0-60% Cu+Cu	K_S^0-h 40-80% Au+Au	K_S^0-h 0-12% Au+Au	Λ-h MB d+Au	Λ-h 0-60% Cu+Cu	Λ-h 40-80% Au+Au	Λ-h 0-12% Au+Au
2.0-2.5	–	0.036 ± 0.004	0.063 ± 0.008	–	–	0.026 ± 0.005	–	–
2.5-3.0	–	0.059 ± 0.006	0.084 ± 0.023	–	–	0.071 ± 0.007	0.061 ± 0.010	–
3.0-3.5		0.098 ± 0.009	0.139 ± 0.022	–		0.084 ± 0.017	0.100 ± 0.019	0.105 ± 0.022
3.5-4.0 4.0-4.5	0.162 ± 0.028	0.144 ± 0.011	0.172 ± 0.021	0.160 ± 0.036	0.079 ± 0.018	0.142 ± 0.013	0.096 ± 0.030	0.128 ± 0.022
4.5-5.0 5.0-5.5	–	–	0.170 ± 0.037	0.240 ± 0.045	–	–	0.184 ± 0.040	0.091 ± 0.033

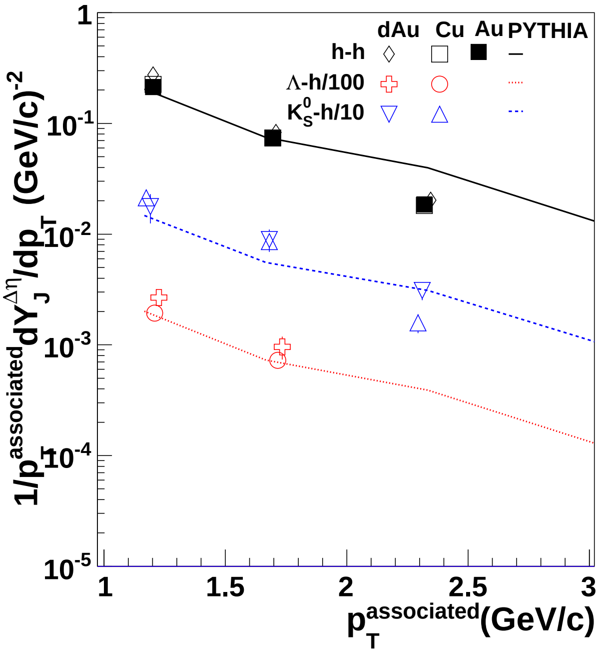

FIG. 5: The jet-like yield as a function of $p_T^{\mathrm{associated}}$ for h-h [7], K^0_S-h, and Λ-h correlations for $3 < p_T^{\mathrm{trigger}} < 6$ GeV/c in d+Au, 0-60% Cu+Cu, and 40-80% Au+Au collsions at $\sqrt{s_{NN}} = 200$ GeV. The data are compared to PYTHIA [40] calculations using the Perugia 2011 tune [41]. Data are binned in $1.0 < p_T^{\mathrm{associated}} < 1.5$ GeV/c, $1.5 < p_T^{\mathrm{associated}} < 2.0$ GeV/c, and $2.0 < p_T^{\mathrm{associated}} < 3.0$ GeV/c and are plotted at the mean of the bin. The 5% systematic error due to the uncertainty on the associated particle's efficiency is not shown.

The jet-like yield as a function of $p_T^{\mathrm{associated}}$ is shown in Figure 5 for h-h [7], K^0_S-h, and Λ-h correlations for d+Au and Cu+Cu collisions at $\sqrt{s_{NN}} = 200$ GeV. All yields are determined using bin counting. The Λ-h and K^0_S-h correlations are only shown for d+Au and Cu+Cu collisions since residual track merging made measurements in Au+Au collisions difficult. These data are compared to PYTHIA [40] calculations from the Perugia 2011 [41] tune. PYTHIA calculations are done in the same bins as the data in order to be directly comparable.

1. *Particle type dependence*

The particle type dependence is of interest because effects such as recombination are expected to be significant in this momentum range for the inclusive spectra, however, the contribution to the jet-like correlation from recombination is expected to be small based on the results in Section IV A. The jet-like yield shows a systematic particle type dependence, with the jet-like yield highest for K^0_S-h correlations and lowest for Λ-h correlations. The fact that this is also observed in d+Au indicates that this particle type dependence is not due to medium modification.

The discrepancy in the particle type ordering between PYTHIA and the data could be from different jet-like correlation widths, differences in the fraction of energy taken by the leading parton, or differences in the shapes of the jet spectra leading to the correlations. The jet-like yield in this paper is defined within a fixed $\Delta\eta$ and $\Delta\phi$ region. In PYTHIA the widths for Λ and K^0_S triggers are the approximately same. It is not possible to extract a reliable width from these data, however, if the jet-like correlation with a Λ trigger were wider than that for the K^0_S trigger, the jet-like yield $Y_J^{\Delta\eta}$ would be lower for the Λ due to the width alone.

The difference in the ordering could also come from an over or underestimate of the fraction of momentum taken by a leading K^0_S or Λ. If the leading parton takes more energy, there is less energy left over for particle production. This means that if PYTHIA underestimates (overestimates) the fraction of momentum taken by a leading Λ (K^0_S), it will overestimate (underestimate) the yield. Similarly, a difference in the jet spectra could lead to a difference in the observed particle ordering. Higher energy jets produce more particles, so if the spectrum of jets leading to the K^0_S-h (Λ-h) correlation is harder (softer) in PYTHIA than in the data, this could explain the discrepancy in the particle typeordering.

The precision of this data set does not allow for studies of the widths of the jet-like yield. However, this would be an area of interest for future studies, particularly in smaller collision systems such as p+p and d+Au. Further information on the fragmentation functions of strange particles in light jets and jets with leading strange particles would also help refine the PYTHIA predictions for the production of strange particles in jets. This could potentially lead to insight into the discrepancies between the data and PYTHIA predictions for the production of baryons and strange particles.

2. *Centrality dependence*

The jet-like yield in h-h collisions as a function of p_T^{trigger}, N_{part}, and $p_T^{\mathrm{associated}}$ is independent of collision system [7], however, the jet-like yield is not independent of collision system in Λ-h and K^0_S-h correlations. Figure 3 and Figure 4 indicate that there are p_T and centrality-dependent modifications to the jet-like correlation in 0-12% central Au+Au collisions. Figure 4 and Figure 5 also indicate that the jet-like yield changes with collision system.

There are several effects which may explain the observed system dependence of the jet-like yield. The basic mechanisms involve either modification of the jet-like correlation itself or differences in the jet spectra which leads to the jet-like correlations. The fragmentation function of the jets could be modified. It is possible that the fragmentation function could be modified for strange particles but not for light quarks. The width of the jet-like correlation could increase, for instance if jets which produce strange particles were more likely to emit bremsstrahlung gluons. Differences in the jet spectra sampled by different trigger particles could also lead to differences in the observed jet-like yield. A higher energy jet produces more particles, so suppression of lower energy jets would lead to an increase in the observed jet-like yield. If identified trigger particles were more likely to come from quark or antiquark jets but unidentified trigger particles were dominated by gluon jets, a difference between quark and gluon jet spectra could lead to differences in the jet-like yield. These data indicate that one or many of these effects are present for correlations with identified strange triggers, however, these data do not have the precision to disentangle these effects.

V. CONCLUSIONS

We observe a trend for mass ordering of the yield of identified strange trigger particles in d+Au, Cu+Cu, and Au+Au collisions. This mass ordering is not the same ordering observed for the jet-like yield in PYTHIA. The $\frac{\Lambda+\bar{\Lambda}}{2K^0_S}$ ratio for the jet-like correlation is comparable to that observed in p+p collisions, providing further evidence that this correlation is dominantly produced by fragmentation. In PYTHIA, this ratio is comparable to the inclusive ratio. The PYTHIA tunes which were investigated are unable to describe either the inclusive ratios or the ratio in the jet-like correlation. This indicates that there may still be some problems with the production mechanism for strange particles in PYTHIA.

These studies provide motivation for future studies of strangeness production in jets. Larger data sets and data from collisions at higher energies could provide more robust tests of the strangeness production mechanism in p+p collisions. Studies in p+p would be essential in order to search for modifications of strangeness production in jets in heavy ion collisions.

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